Main Wheel (Mod S.O.O. 6048 and Mod 6/2183) – General data

1. General

On aircraft with Mod S.O.O. 6048 and Mod 6/2183 incorporated the main wheel hub is fitted with a 36 x 13 x 12 tire, a brake spacer installed between the hub and brake disc, a valve stem bushing and the wheel assembly bolts replaced by longer bolts to accommodate the spacer.

2. Removal/Installation

A. Remove Main Wheel

1) Remove the brake assembly and place it on a clean flat surface, as per 32-40-31, Wheel Brakes Unit (Mod 6/2002) – General Data.

2) Jack aircraft until wheel is clear of ground. (Refer to 07-00-00, Lifting and Shoring – General Data).

3) Deflate tire.

4) Remove snap ring and hubcap.

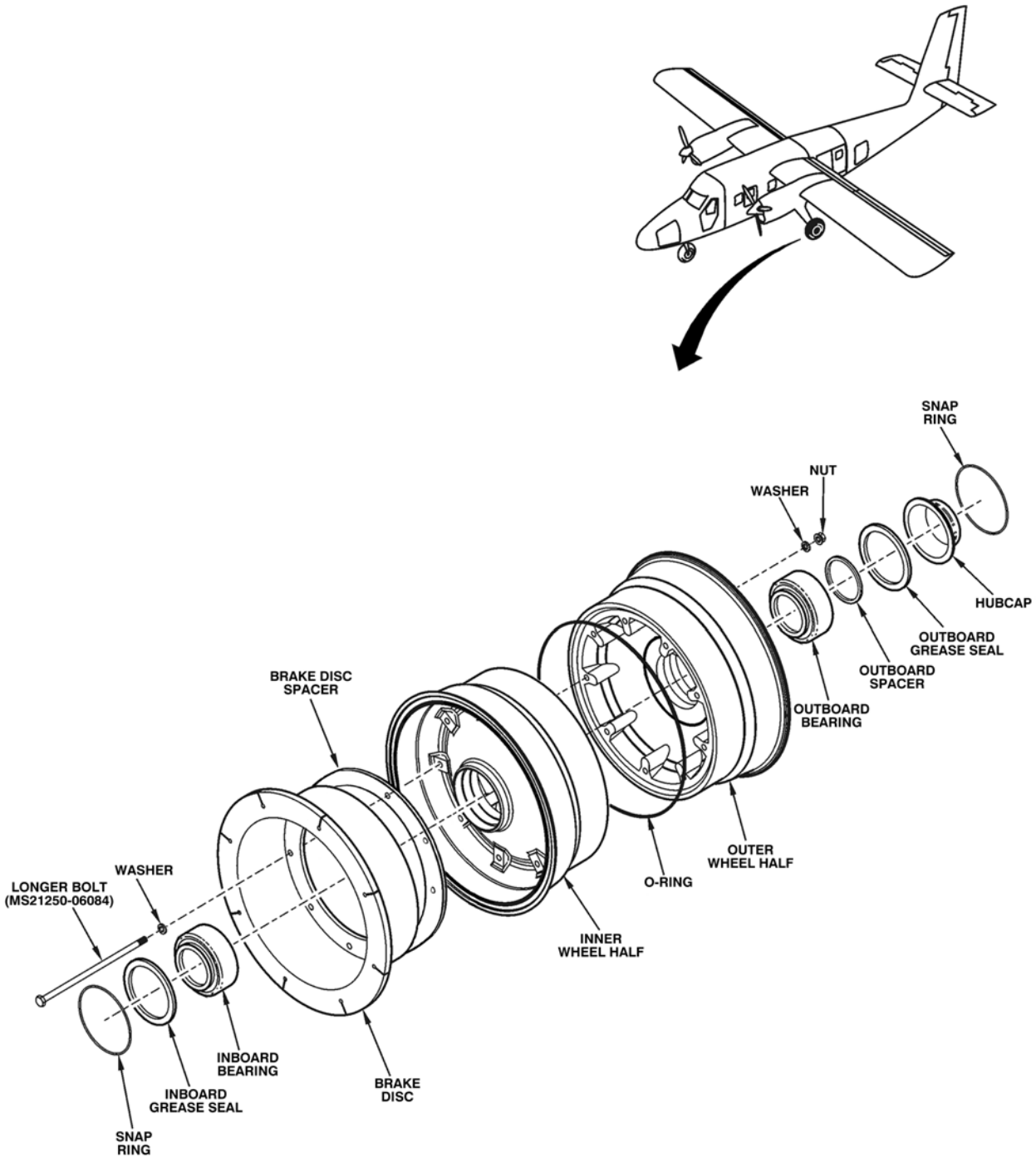
5) Remove two safety bolts and self locking nuts from axle nut. Remove axle nut using wrench SD5523.

6) Remove outboard grease seal, spacer and bearing.

Note Place wheel seals, spacer and bearings on a clean, flat surface to prevent any contamination.

7) Carefully slide wheel/tire assembly off of axle.

8) Remove inboard bearing, seal and retaining clip.



Main Wheel Assembly (Mod S.O.O. 6048 Mod 6/2183)
Figure 1

B. Install Main Wheel

1) Apply a thin coat of grease (MIL–PRF–81322) to axle and axle threads.

2) Install lubricated inboard bearing in wheel assembly with grease seal and retaining clip.

3) Carefully slide wheel/tire assembly onto axle.

4) Install lubricated outboard bearing into wheel assembly with spacer and grease seal.

a) Install axle nut using wrench SD5523 and torque to 150 in–lbs to 200 in–lbs while seating the bearing.

b) Back off axle nut.

c) Retighten to 30 in–lbs to 40 in–lbs while rotating.

d) Rotate axle nut to nearest axle slot and install two safety bolts and self locking nuts.

Note Wheel must rotate freely, without perceptible play.

5) Install hub cap and snap ring.

6) Inflate tire.

7) Reinstall brake assembly as per 32-40-31, Wheel Brakes Unit (Mod 6/2002) – General Data.

Note If installation of main wheel on aircraft is part of gear configuration change then do step as per Para 2.B.8).

8) Do procedure for gear configuration change as per 32-10-11, Main Gear Leg – General Data, Para 2.C.

C. Remove Main Wheel Tire (with tube) (Mod S.O.O. 6048 and Mod 6/2183)

1) Remove the wheel/tire assembly from aircraft as per Para 2.A.

2) Release all pressure from tire by slackening air valve core.

3) Remove the valve stem bushing from wheel assy and valve core.

4) Remove valve core completely from wheel assy.

5) Using tire demounters, wooden wedges, or an approved tire machine, separate edges of tire from wheel hub beading.

6) Remove bolts, self-locking nuts, washers and countersunk washers, holding the brake disc, brake disc spacer and halves of the hub together. Refer to Figure 1.

7) Carefully separate hub halves, and remove O-ring and tire. Check O-ring for serviceability and discard if damaged.

D. Install Main Wheel Tire (with tube) (Mod S.O.O. 6048 and Mod 6/2183)

1) Check hub for scratches, scores and cleanliness according to OVH Manual 040–21101. Clean tire area of hub with a dry cleaning solvent. Dust area with French chalk (talc).

2) Partially inflate tube and position in tire. Align yellow stripes on the tube base with the red dot on the tire sidewall.

3) Position hub half containing air valve hole on the bench and place serviceable tire and O-ring on the hub with red spot in tire aligned with control valve. Install position valve stem through valve hole and secure with the valve stem bushing. Ensure valve stem bushing is snug.

CAUTION **ENSURE BOLTS ARE MS21250–06084. DO NOT USE BOLTS THAT ARE PROVIDED WITH WHEEL ASSEMBLY FOR STANDARD GEAR, (MS21250–06080).**

4) Lubricate threads of bolts and bearing surfaces of nuts, bolt heads and washers with anti-seize compund (MIL–T–5544A).

5) Install O-ring, other half hub, brake disc spacer and brake disc. Install bolts and washers and self locking nuts, then tighten by equal increments in diagonal order to torque values stamped on wheel hub.

Note Install bolts with heads on inboard side of wheel with one washer under nut and one under bolt head.

WARNING **DO NOT INFLATE TIRE UNTIL ALL BOLTS HAVE BEEN PROPERLY TIGHTENED. IT IS RECOMMENDED THAT ALL MOUNTED TIRES BE INFLATED INSIDE A METAL CAGE TO AVOID POSSIBLE INJURY TO PERSONNEL.**

CAUTION **DO NOT USE IMPACT OR POWER WRENCHES TO TIGHTEN OR TORQUE WHEEL BOLTS OR NUTS.**

6) Inflate tire to correct pressure. (Refer to 12-10-20, Landing Gear – General Data).

7) Apply a 1” x 3” creep mark using a suitable contrasting coloured paint between tire and rim.

8) Install wheel on aircraft as per Para 2.B.

3. Inspection/Check

A. Inspect Main Wheel Hub

1) Remove main wheel and main wheel tire. (Refer to Para 2.A and Para 2.C).

2) Clean hub halves with dry cleaning solvent (P–D–680 or equivalent) and dry with filtered and dried compressed air. Refer to OVH Manual 040–21101.

3) Inspect hub halves for cracks (using the fluorescent penetrant method or an approved dye penetrant method), chipping of rims, damage and corrosion. Replace hub if cracks or chipping of rims is evident. Refer to OVH Manual 040–21101.

Note Small nicks or gouges can be blended out and polished with fine aluminum oxide cloth (refer to Manufacturer's Overhaul Manual for definition and limitation of damage). Refer to OHV Manual 040–21101.

4) Inspect bearings and replace if damaged. Refer to OVH Manual 040–21101.

5) Inspect brake disc spacer for damage or corrosion. Replace if damaged.

6) Inspect brake disc wheel bolts and nuts as per OVH 030–17000.

7) Install tire to main wheel and install main wheel on aircraft. (Refer to Para 2.B and Para 2.D).